

Calibration of Local Volatility Models

A Tikhonov regularization approach to the Dupire inverse problem

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Introduction

- ▶ An option gives its holder the right, but not the obligation, to buy (call) or sell (put) an asset at an agreed price K on an agreed date T .
- ▶ In 1973 Black, Scholes and Merton priced European options with a PDE whose solution has a closed form, assuming a **constant volatility** σ .
- ▶ After the 1987 crash markets display the **smile**: the volatility implied by the prices depends on the strike and the maturity.
- ▶ Dupire (1994) proposed the **local volatility model**, where the volatility is a deterministic function $\sigma_{\text{loc}}(t, S)$, which reproduces any arbitrage-free set of prices.
- ▶ To be useful the model must be **calibrated**: find the coefficient σ_{loc} of a PDE from finitely many, noisy observations of its solution. This is an ill-posed **inverse problem**.

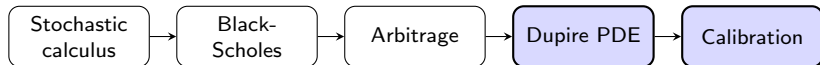
Aim: build the pricing theory up to the Dupire equation, and solve its calibration by Tikhonov regularization following

Egger, H., Engl, H. W. (2005). *Tikhonov regularization applied to the inverse problem of option pricing: convergence analysis and rates*. Inverse Problems, 21(3), 1027.

All the code can be found in <https://github.com/M315/TFM>.

Structure of the work

- 2–3. **Preliminaries.** Probability spaces, filtrations, martingales, Brownian motion, Itô integral, Itô formula and SDEs.
 - 4. **Black-Scholes.** Replication, the BS PDE, its solution, implied volatility and its surface.
 - 5. **Arbitrage theory.** PDE and martingale methods, fundamental theorems of asset pricing.
 - 6. **Local volatility.** Dupire formula, forward equation, no-arbitrage conditions, local vs. implied volatility.
 - 7. **Calibration.** Ill-posedness, Tikhonov functional, adjoint gradient, convergence rates, discretization and numerical experiments.
- A–C. **Appendices.** Implied volatility pipeline, calibration details, code.



This presentation briefly recalls stochastic calculus and Black-Scholes, and focuses on the Dupire PDE and its calibration.

Stochastic calculus. Itô integral and Itô's formula

- ▶ An ODE $dS_t = f(t, S_t) dt$ cannot capture the random fluctuations of prices. A **stochastic differential equation** adds a Brownian noise term,

$$dS_t = b(t, S_t) dt + \sigma(t, S_t) dW_t \iff S_t = S_0 + \int_0^t b(s, S_s) ds + \int_0^t \sigma(s, S_s) dW_s.$$

- ▶ **Itô integral**: for elementary $\phi = \sum_j e_j \mathbf{1}_{[t_j, t_{j+1})}$, with e_j $\mathcal{F}_{t_j}$ -measurable,

$$\int_0^T \phi dW_t = \sum_j e_j (W_{t_{j+1}} - W_{t_j}), \quad \mathbb{E} \left[\left(\int_0^T \phi dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T \phi^2 dt \right],$$

the second identity being the **Itô isometry**. It is extended by density to adapted processes with $\mathbb{E} \int_0^T f^2 dt < \infty$. The result is a continuous **martingale**.

- ▶ **Itô's formula** (stochastic chain rule): if $dX_t = \mu dt + \sigma dW_t$ and $g \in C^{1,2}$, then $Y_t = g(t, X_t)$ is an Itô process with

$$dY_t = \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} (dX_t)^2, \quad (dW_t)^2 = dt, \quad dt dW_t = (dt)^2 = 0.$$

- ▶ The extra second-order term $\frac{1}{2} \sigma^2 g_{xx} dt$ is what turns stochastic dynamics into the **diffusion term of a PDE**.

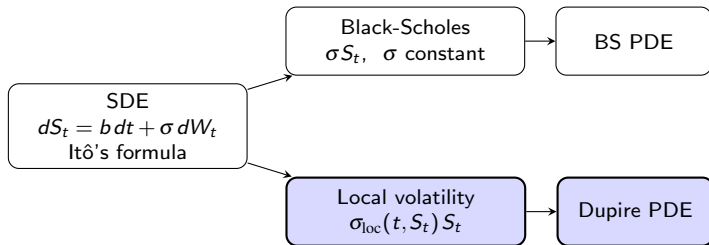
Stochastic calculus. SDEs as building blocks

- **Existence and uniqueness:** if b and σ are Lipschitz in x and have linear growth, the SDE has a unique continuous adapted solution.
- **Geometric Brownian motion.** Applying Itô's formula to $\ln S_t$,

$$dS_t = \mu S_t dt + \sigma S_t dW_t \implies d(\ln S_t) = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW_t,$$

so $S_t = S_0 \exp[(\mu - \sigma^2/2)t + \sigma W_t]$. Log-normal prices, always positive: the Black-Scholes model.

- Each model of this work is an SDE for the underlying, differing only in the diffusion coefficient:



Black-Scholes. Dynamic replication

- ▶ Arbitrage-free market, constant rate r , continuous rebalancing. The underlying and the bond follow

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad d\beta_t = r\beta_t dt.$$

- ▶ Replicating portfolio $V_t = a_t S_t + b_t \beta_t$ with $V_T = (S_T - K)^+$, and **self-financing**

$$dV_t = a_t dS_t + b_t d\beta_t.$$

- ▶ Writing $V_t = f(t, S_t)$, applying Itô's formula and matching the dW_t and dt terms,

$$a_t = \frac{\partial f}{\partial S} \quad (\text{delta hedge}),$$

which gives the **Black-Scholes PDE**

$$\frac{\partial f}{\partial t} = rf - rS \frac{\partial f}{\partial S} - \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 f}{\partial S^2}, \quad f(T, S) = (S - K)^+.$$

- ▶ The drift μ does not appear: the price does not depend on the investors' view of the asset.

Black, F., Scholes, M. (1973). *The Pricing of Options and Corporate Liabilities*. J. Political Economy.

Steele, J. M. (2010). *Stochastic Calculus and Financial Applications*. Springer.

Black-Scholes. Pricing formula

- ▶ With $\tau = T - t$, log-moneyness $x = \ln(S/K)$ and $u = e^{\alpha x + \beta \tau} v$ for suitable α, β , the BS PDE reduces to the **heat equation**

$$v_\tau = \frac{1}{2} \sigma^2 v_{xx}, \quad v(x, 0) = Ke^{-\alpha x} (e^x - 1)^+.$$

- ▶ Convolving with the heat kernel and completing the square gives the **Black-Scholes formula**

$$f(t, S) = S \Phi(d_1) - Ke^{-r(T-t)} \Phi(d_2),$$

$$d_{1,2} = \frac{\ln(S/K) + (r \pm \frac{1}{2} \sigma^2)(T-t)}{\sigma \sqrt{T-t}}.$$

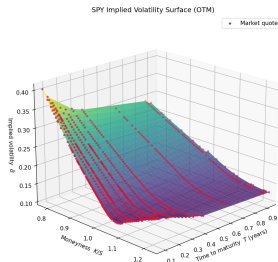
- ▶ **Implied volatility:** the unique $\sigma_{iv}(K, T) > 0$ such that $C_{BS}(\sigma_{iv}) = C^{mkt}$. Uniqueness follows from the vega

$$\gamma = \frac{\partial C_{BS}}{\partial \sigma} = S \sqrt{T-t} \phi(d_1) > 0,$$

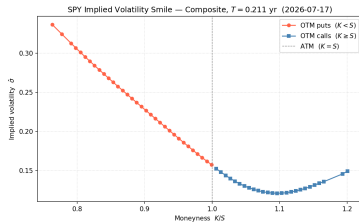
and it is computed with Newton's method
 $\sigma^{(n+1)} = \sigma^{(n)} - (C_{BS}(\sigma^{(n)}) - C^{mkt}) / \gamma(\sigma^{(n)})$.

Black-Scholes. The implied volatility surface

If Black-Scholes described the market, $\sigma_{iv}(K, T)$ would be flat. SPY quotes on 30 April 2026 show a **negative skew** in strike and a **term structure** in maturity.



(a) Implied volatility surface.



(b) Smile at $T \approx 0.21$.

Constant σ is inconsistent with the market, which motivates the local volatility model.

Arbitrage theory. Market, portfolios and arbitrage

- **Market:** a riskless asset $dS_0 = rS_0 dt$ and risky assets

$$dS_i(t) = \mu_i(S(t), t) dt + \sigma_i(S(t), t)^\top dW(t), \quad i = 1, \dots, n.$$

- **Portfolio:** an adapted process $\theta(t)$ with value $V(t) = \theta(t)^\top S(t)$, self-financing if $dV(t) = \theta(t)^\top dS(t)$.
- **PDE method:** replicating $V(S(t), t)$ and imposing self-financing gives $\theta_i = \partial V / \partial S_i$ and

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{i,j} \frac{\partial^2 V}{\partial S_i \partial S_j} \sigma_i^\top \sigma_j = 0, \quad V = \sum_i \frac{\partial V}{\partial S_i} S_i,$$

which recovers the BS PDE for $n = 1$, $\sigma_1 = \sigma S_1$.

- **Arbitrage:** a self-financing admissible portfolio with
 1. $V(0) < 0$ and $\mathbb{P}(V(t) \geq 0) = 1$, or
 2. $V(0) = 0$, $\mathbb{P}(V(t) \geq 0) = 1$ and $\mathbb{P}(V(t) > 0) > 0$.

Björk, T. (2009). *Arbitrage Theory in Continuous Time*. Oxford University Press.

Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering*. Springer.

Arbitrage theory. Martingale method

- ▶ A **risk-neutral measure** $\mathbb{Q} \sim \mathbb{P}$ is one under which the normalized prices $Z_i = S_i/S_0$ are martingales: every asset drifts at the risk-free rate.
- ▶ **First fundamental theorem.** The market is arbitrage-free if and only if there exists an equivalent (local) martingale measure $\mathbb{Q}$.
- ▶ **Second fundamental theorem.** Assuming no arbitrage, the market is complete if and only if $\mathbb{Q}$ is unique.
- ▶ In a complete market the price of a claim with payoff $f(S(T))$ is

$$V(t) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(s) ds} f(S(T)) \middle| \mathcal{F}_t \right].$$

For a call with constant r ,

$$C(K, T) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+],$$

which is the starting point of the Dupire equation.

Local volatility model

- ▶ The volatility becomes a deterministic function of time and spot,

$$\frac{dS_t}{S_t} = (r - q)dt + \sigma_{\text{loc}}(t, S_t)dW_t,$$

with q the dividend yield. We write σ_{loc} for the local and σ_{iv} for the implied volatility.

- ▶ The pricing PDE is the Black-Scholes one with σ replaced by $\sigma_{\text{loc}}(t, S)$,

$$\frac{\partial f}{\partial t} = rf - (r - q)S \frac{\partial f}{\partial S} - \frac{1}{2} \sigma_{\text{loc}}^2(t, S) S^2 \frac{\partial^2 f}{\partial S^2}.$$

- ▶ It is a **market model**: it is calibrated so that vanilla options are priced exactly.
- ▶ **Question**: given the call prices $C(K, T)$, which σ_{loc} produces them?

Dupire, B. (1994). *Pricing with a Smile*. Risk.

Bergomi, L. (2015). *Stochastic Volatility Modeling*, Ch. 2. CRC Press.

Dupire equation. Derivation (I)

Take a general diffusion $dS_t = (r - q)S_t dt + \sigma_t S_t dW_t$, σ_t any adapted process, and undiscounted prices $\mathcal{C}(K, T) = e^{rT} C(K, T) = \mathbb{E}[(S_T - K)^+]$.

- In the sense of distributions, with θ the Heaviside function and δ the Dirac delta,

$$\frac{d}{dx}(x)^+ = \theta(x), \quad \frac{d^2}{dx^2}(x)^+ = \delta(x).$$

- Itô's formula applied to $(S_T - K)^+$ over $[T, T + dT]$,

$$d(S_T - K)^+ = \theta(S_T - K)((r - q)S_T dT + \sigma_T S_T dW_T) + \frac{1}{2} \delta(S_T - K) \sigma_T^2 S_T^2 dT.$$

- Differentiating $\mathcal{C}$ in K ,

$$\mathbb{E}[\theta(S_T - K)] = -\frac{\partial \mathcal{C}}{\partial K}, \quad \mathbb{E}[\delta(S_T - K)] = \frac{\partial^2 \mathcal{C}}{\partial K^2} = p^*(K, T),$$

the second one is **Breeden-Litzenberger**: the convexity in strike is the risk-neutral density of S_T .

Breeden, D. T., Litzenberger, R. H. (1978). *Prices of State-Contingent Claims Implicit in Option Prices*. J. Business.

Dupire equation. Derivation (II)

- ▶ Taking expectations ($\mathbb{E}[dW_T] = 0$) and using the sifting property $S_T^2 \delta(S_T - K) = K^2 \delta(S_T - K)$,

$$\frac{\partial \mathcal{C}}{\partial T} = (r - q) \left(\mathcal{C} - K \frac{\partial \mathcal{C}}{\partial K} \right) + \frac{K^2}{2} \mathbb{E}[\sigma_T^2 \delta(S_T - K)].$$

- ▶ Dividing by $\mathbb{E}[\delta(S_T - K)]$ and going back to discounted prices,

$$\mathbb{E}[\sigma_T^2 | S_T = K] = 2 \frac{\frac{\partial C}{\partial T} + qC + (r - q)K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}},$$

valid for *any* σ_t : the smile only fixes the conditional expectation of σ_T^2 .

- ▶ In the local volatility model $\sigma_t = \sigma_{\text{loc}}(t, S_t)$, which gives the **Dupire formula**

$$\sigma_{\text{loc}}(t, S)^2 = 2 \frac{\frac{\partial C}{\partial T} + qC + (r - q)K \frac{\partial C}{\partial K}}{K^2 \frac{\partial^2 C}{\partial K^2}} \bigg|_{\substack{K=S, \\ T=t}}.$$

- ▶ A full surface of call prices determines σ_{loc} uniquely.

Dupire forward equation

Rearranging the same identity, for a known σ_{loc} the call prices solve the **Dupire forward equation**

$$\frac{\partial C}{\partial T} + (r - q)K \frac{\partial C}{\partial K} - \frac{\sigma_{\text{loc}}^2(T, K)}{2} K^2 \frac{\partial^2 C}{\partial K^2} = -qC, \quad (1)$$

$$(K, T) \in (0, \infty) \times (0, T_{\max}], \quad C(K, 0) = (S_0 - K)^+.$$

	Black-Scholes PDE	Dupire PDE
Variables	spot S , time t	strike K , maturity T
Direction	backward from payoff at T	forward from $T = 0$
Fixed	one strike and maturity	one spot S_0
Output	price for every S	prices for every (K, T)

- One solve prices the whole surface quoted today. This is what makes the equation the natural **direct problem** for calibration: the data $C^*(K_i, T_j)$ are values of its solution, and the unknown is its coefficient σ_{loc} .

No-arbitrage conditions

No arbitrage makes numerator and denominator of the Dupire formula non-negative, so $\sigma_{\text{loc}}^2 \geq 0$ wherever the density of S_T does not vanish.

- **Strike (butterfly) arbitrage.** $\partial^2 C / \partial K^2 = e^{-rT} p^*(K, T) \geq 0$. Otherwise the butterfly

$$V_0(\varepsilon) = \frac{C(K - \varepsilon, T) - 2C(K, T) + C(K + \varepsilon, T)}{\varepsilon^2} \longrightarrow \frac{\partial^2 C}{\partial K^2} < 0$$

is entered at a negative premium with a non-negative payoff, by convexity of $(x)^+$.

- **Maturity (calendar) arbitrage.** The numerator equals

$$e^{-qT} \frac{d}{dT} \left[e^{qT} C \left(K e^{(r-q)T}, T \right) \right] \geq 0,$$

a call struck at a fixed proportion of the forward cannot get cheaper with maturity.

Both are conditions on the *prices*: every arbitrage-free surface admits a local volatility.

From local to implied volatility

- **Gamma average.** Delta hedging with σ_{iv} while the spot follows σ_{loc} gives

$$\sigma_{iv}(K, T)^2 = \frac{\mathbb{E} \left[\int_0^T e^{-rt} S_t^2 \partial_{SS} C_{BS} \sigma_{loc}(t, S_t)^2 dt \right]}{\mathbb{E} \left[\int_0^T e^{-rt} S_t^2 \partial_{SS} C_{BS} dt \right]} :$$

the implied variance is a smoothed local variance, so the smile is flatter.

- **Rule of two.** If $\sigma_{loc}(t, S) = \sigma_0 + \alpha(t)x + \dots$, $x = \ln(S/F_t)$, the ATMF skew $\mathcal{S}_T = \partial_{\ln K} \sigma_{iv}|_{K=F_T}$ is, to first order,

$$\mathcal{S}_T = \frac{1}{T} \int_0^T \frac{t}{T} \alpha(t) dt \stackrel{\alpha \text{ const.}}{\implies} \mathcal{S}_T = \frac{\alpha}{2}.$$

The local skew is twice the implied skew.

- **Skew stickiness ratio.** $\mathcal{R}_T = \frac{1}{\mathcal{S}_T} \frac{d\sigma_{ATMF}}{d \ln S_0} = 1 + \frac{1}{T} \int_0^T \frac{\mathcal{S}_t}{\mathcal{S}_T} dt$, equal to 2 for a flat term structure of skew: the smile dynamics are not free, they are fixed by the calibrated smile.

Bergomi, L. (2015). *Stochastic Volatility Modeling*, Ch. 2. CRC Press.

Derman, E., Miller, M. B., Park, D. (2016). *The Volatility Smile*. Wiley.

Calibration. An ill-posed inverse problem

Data: quoted calls $C^*(K_i, T_j)$ on finitely many strikes and maturities. **Unknown:** the coefficient σ_{loc} of the forward equation (1).

Why not plug the quotes into the Dupire formula?

- ▶ The data are **noisy** and **sparse**, especially in maturity, and the formula needs $\partial_T C$ and $\partial_{KK} C$.
- ▶ Gamma $\partial_{KK} C$ sits in the denominator: far from the money it is small and σ_{loc} is barely determined by the prices.
- ▶ The map from σ_{loc} to prices is **smoothing**: very different volatilities produce almost the same prices, so its inverse is discontinuous and amplifies the noise.

We follow Egger and Engl and stabilize the inversion by **Tikhonov regularization**.

Egger, H., Engl, H. W. (2005). *Tikhonov regularization applied to the inverse problem of option pricing: convergence analysis and rates*. Inverse Problems.

Engl, H. W., Hanke, M., Neubauer, A. (1996). *Regularization of Inverse Problems*. Kluwer.

Calibration. Log-moneyness formulation

With $y = \log(K/S_0)$, $\tau = T$, $u(y, \tau) = C/S_0$ and the diffusion coefficient

$$a(y, \tau) = \frac{1}{2} \sigma_{\text{loc}}^2(T, K),$$

the operators $K\partial_K$ and $K^2\partial_{KK}$ have constant coefficients,

$$K \frac{\partial C}{\partial K} = S_0 u_y, \quad K^2 \frac{\partial^2 C}{\partial K^2} = S_0 (u_{yy} - u_y),$$

and the forward equation becomes, on $Q = (-M, M) \times (0, T]$,

$$u_\tau = a(y, \tau) (u_{yy} - u_y) - (r - q) u_y - q u, \quad (2)$$

$$u(y, 0) = (1 - e^y)^+, \quad u(-M, \tau) = \Psi_1(\tau), \quad u(M, \tau) = \Psi_2(\tau),$$

with the boundary values from Black-Scholes at $K = S_0 e^{\pm M}$ ($\Psi_1 \rightarrow e^{-q\tau}$, $\Psi_2 \rightarrow 0$ as $M \rightarrow \infty$).

Calibration. Parameter-to-solution map

- ▶ Admissible set, for a prior a^* and bounds $0 < \underline{a} \leq \bar{a}$,

$$\mathcal{K}(a^*) := \{a \in a^* + H^1(Q) : \underline{a} \leq a \leq \bar{a}\}.$$

H^1 is needed because a enters the weak form through the flux, and its norm gives weak compactness of the sublevel sets.

- ▶ Parameter-to-solution map, $u(a)$ the solution of (2),

$$F(a) = u(a) - u(a_0), \quad a_0 \text{ constant, } u(a_0) \text{ from Black-Scholes.}$$

Theorem (Egger and Engl, Thm. 2.1)

There exists $p^* > 2$ such that $F: \mathcal{K}(a^*) \rightarrow W_p^{2,1}(Q)$ is continuous and compact for $2 \leq p < p^*$, and weakly sequentially continuous.

- ▶ This is the **source of the ill-posedness**: a compact operator between infinite dimensional spaces has no continuous inverse, small changes in prices may correspond to order one changes in a .

Calibration. Tikhonov functional

Assume $a^\dagger \in \mathcal{K}(a^*)$ reproduces the true prices $u^\dagger$ and the observations satisfy $\|u^\delta - u^\dagger\|_{L^2(Q)} \leq \delta$. We minimize

$$J^\beta(a) = \underbrace{\|u(a) - u^\delta\|_{L^2(Q)}^2}_{\text{data fit}} + \beta \underbrace{\|a - a^*\|_{H^1(\Omega)}^2}_{\text{smoothness, prior}}, \quad \Omega = (-M, M).$$

Proposition (Egger and Engl, Props. 3.1 and 3.2)

For any $\beta > 0$ a minimizer a_β^δ exists. If $\delta \rightarrow 0$, $\beta(\delta) \rightarrow 0$ and $\delta^2/\beta(\delta) \rightarrow 0$, the minimizers have a subsequence converging in $H^1(\Omega)$, and every limit point is an a^* -minimum-norm solution.

- ▶ The prior a^* carries the structural knowledge where the data are weak.
- ▶ With finitely many quotes the data are discrete: convergence to an a^* -minimum-norm solution holds, its uniqueness does not.

Calibration. Gradient by the adjoint method

- ▶ The directional derivative needs the linearized solution $w = u'(a)[h]$,

$$\frac{d}{d\varepsilon} \|u(a + \varepsilon h) - u^\delta\|^2 \Big|_{\varepsilon=0} = 2(u(a) - u^\delta, w),$$

$$-w_\tau + a(w_{yy} - w_y) - (r - q)w_y - qw = -h(u_{yy} - u_y),$$

one PDE solve per direction h , i.e. n solves for n degrees of freedom.

- ▶ Instead solve once the **adjoint** problem backward in τ ,

$$V_\tau + (aV)_{yy} + (aV)_y + (r - q)V_y - qV = 0, \quad V(y, T) = u(a)(y, T) - u^\delta(y),$$

with homogeneous Dirichlet conditions. Then

$$\langle \nabla J^\beta(a), h \rangle = 2 \int_0^T \int_{-M}^M h(u_{yy} - u_y) V dy d\tau + 2\beta \int_{-M}^M [(a - a^*)h + \partial_y(a - a^*)\partial_y h] dy.$$

- ▶ Cost: **one forward and one adjoint solve per iteration**, independent of the number of unknowns.
- ▶ V represents $F'(a)^*(F(a) - u^\delta)$, the object appearing in the source condition.

Calibration. Convergence rates

- Under the **source condition**

$$a^* - a^\dagger = F'(a^\dagger)^* w, \quad \|w\| \text{ small,}$$

and the choice $\beta \sim \delta$ (Egger and Engl, Prop. 4.1),

$$\|a_\beta^\delta - a^\dagger\| = O(\sqrt{\delta}), \quad \|u(a_\beta^\delta) - u^\delta\| = O(\delta).$$

- For time independent $a = a(y)$ and one maturity, the condition is implied by smooth source functions g with Gaussian decay in the wings

$$|g|, |\partial_y g|, |\partial_y^2 g| = O(e^{-B'y^2}), \quad |y| \rightarrow \infty,$$

with an H^2 penalty (Thm. 4.1) or with observed digital prices C_K (Thm. 4.2).

- **Interpretation:** $a^\dagger - a^*$ must be small and smooth in the wings, where the prices carry little information. The wings must come from the prior.

Caveat: we keep the H^1 penalty with price data only, and later calibrate a time dependent $a(y, \tau)$, both outside the rate theorems.

Calibration. Discretization and minimization

- **Space:** P_1 Lagrange finite elements on $[-M, M]$, with FEniCSx. Weak form, $v \in H_0^1$,

$$\frac{d}{d\tau} \int u v = - \int a u_y v_y - \int \partial_y a u_y v - \int (a + r - q) u_y v - \int q u v.$$

- **Time:** implicit Euler, unconditionally stable for this parabolic operator.
- **Optimizer:** L-BFGS-B, which handles the bounds $\underline{a} \leq a \leq \bar{a}$.
- **Choice of β :** decreasing sequence $\beta_0 = 10^2 \delta > \dots > 10^{-3} \delta$, warm started, stopped by the Morozov discrepancy principle $\|u(a_\beta^\delta) - u^\delta\| \leq 1.5 \delta$.

Gradient check. Adjoint vs. linearized solve agree to 2.3×10^{-15} , and the Taylor remainder $R_1(t) = |J(a + th) - J(a) - t \langle \nabla J, h \rangle|$ is $O(t^2)$:

t	$R_0(t)$	$R_1(t)$	rate of R_1
10^{-2}	1.32×10^{-6}	5.13×10^{-6}	-
10^{-3}	3.28×10^{-7}	5.27×10^{-8}	1.99
10^{-4}	3.76×10^{-8}	5.29×10^{-10}	2.00
10^{-5}	3.80×10^{-9}	5.29×10^{-12}	2.00

Example 1. Constant volatility

Egger and Engl, Example 1: recover $a^\dagger = 0.15$ ($\sigma_{\text{loc}} \approx 0.548$) on $y \in [-3, 3]$, $\tau \in [0, 1]$, $r = q = 0$, $S_0 = 1000$, prior $a^*(y) = 0.15 - 0.05 \operatorname{erf}(-y^2)$, $\beta = 10^{-6}$. Cases: (A) exact data, (B) 20 strikes, (C) poor prior $a^* \equiv 0.15$, (D) 0.1% noise, (BD) both.

Strike	True	A	B	C	D	BD
600	0.1500	0.1493	0.1500	0.1519	0.1378	0.1245
1000	0.1500	0.1499	0.1499	0.1515	0.1490	0.1455
1400	0.1500	0.1497	0.1498	0.1504	0.1489	0.1522
1600	0.1500	0.1508	0.1501	0.1491	0.1543	0.1500
1800	0.1500	0.1487	0.1504	0.1511	0.1461	0.1463

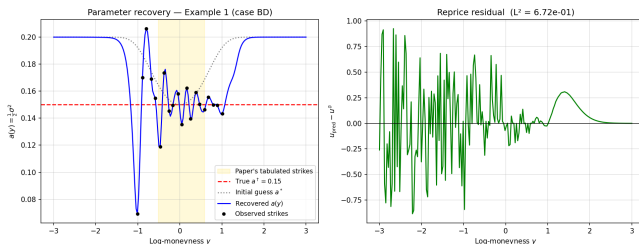


Figure: Case BD: incomplete and noisy data. Good recovery where observed, prior dominates the wings.

Example 2. The role of the prior

Oscillatory true coefficient near the money,

$$a^\dagger(y) = 0.15 + 0.05 e^{-(y+0.3)^2} \sin(2\pi y) + 0.05 \operatorname{erf}(20y), \text{ prior}$$

$a^*(y) = 0.15 + 0.05 \operatorname{erf}(2y)$. Data generated on a finer mesh and wider domain to

avoid the *inverse crime*; noisy cases use the β schedule with the discrepancy principle.

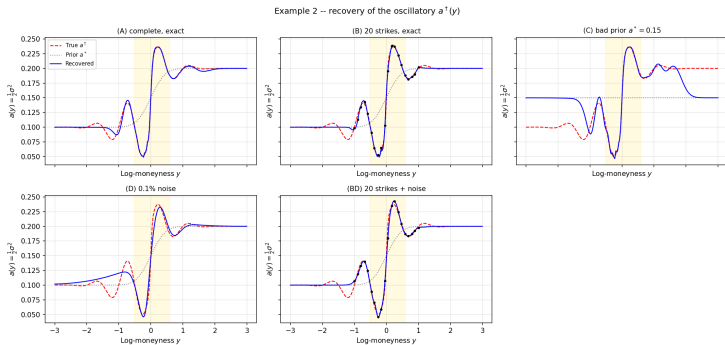


Figure: Recovered $a(y)$ (blue), true $a^\dagger$ (red dashed), prior (grey). In case C the constant prior cannot recover the wings, where the data are weak.

Market surface. Extension to time-dependent volatility

- Prices are observed at finitely many maturities, so a is taken **piecewise constant in time**,

$$a(y, \tau) = \sum_{n=1}^K a_n(y) \chi_{(\tau_{n-1}, \tau_n]}(\tau), \quad a_n \in H^1(-M, M),$$

one slice per quoted maturity. $K = 1$ is the setting of Egger and Engl.

- The functional adds a penalty on jumps between slices,

$$J(a) = \sum_{j=1}^N \|u(a)(\cdot, \tau_j) - u_j^\delta\|_{L^2(\Omega_j)}^2 + \beta_y \sum_{n=1}^K \|a_n - a_n^*\|_{H^1(\Omega)}^2 + \beta_\tau \sum_{n=2}^K \|a_n - a_{n-1}\|_{L^2(\Omega)}^2.$$

One forward march to the last maturity and one adjoint sweep give the whole gradient.

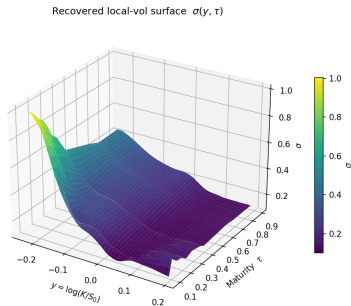
- Prior.** Seeding with $\frac{1}{2} \sigma_{iv}^2$ anchors the slope at half the right value (rule of two). Instead invert the short maturity harmonic mean relation of Berestycki, Busca and Florent,

$$\frac{y}{\sigma_{iv}(y)} = \int_0^y \frac{dx}{\sigma_{loc}(x)} \implies \sigma_{loc}(y) = \frac{\sigma_{iv}(y)}{1 - y \partial_y \sigma_{iv}(y) / \sigma_{iv}(y)}.$$

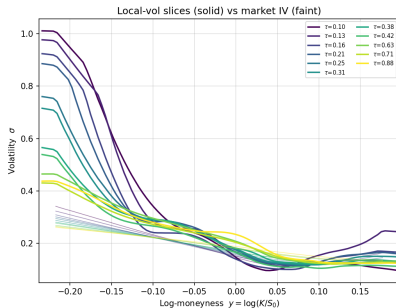
Berestycki, H., Busca, J., Florent, I. (2002). *Asymptotics and calibration of local volatility models*. Quantitative Finance.

Market surface. SPY, 30 April 2026

$S_0 = 720.65$, eleven maturities $\tau \in [0.10, 0.88]$, liquid band $0.8 \leq K/S_0 \leq 1.2$, $\beta_y = 0.1$, $\beta_\tau = 1$.



(a) Calibrated $\sigma_{\text{loc}}(y, \tau)$.



(b) Local (solid) vs. implied (faint) volatility.

Pronounced negative skew: $\sigma_{\text{loc}} \approx 0.15$ ATM at the short end to 0.23 at the long end, up to 0.95 on the short downside. Median reprice error 0.13%, below 1% on liquid strikes at every maturity (the naive prior $\frac{1}{2} \sigma_{\text{iv}}^2$ gives 0.20% and maxima above 2%).

Market surface. Free of arbitrage by construction

- ▶ The optimizer keeps $a \geq 10^{-8}$ and the prices solve the Dupire equation with that coefficient, so they are the prices of a genuine diffusion: the conditions of the no-arbitrage slide cannot fail.
- ▶ The implied volatility interpolation route carries no such guarantee.

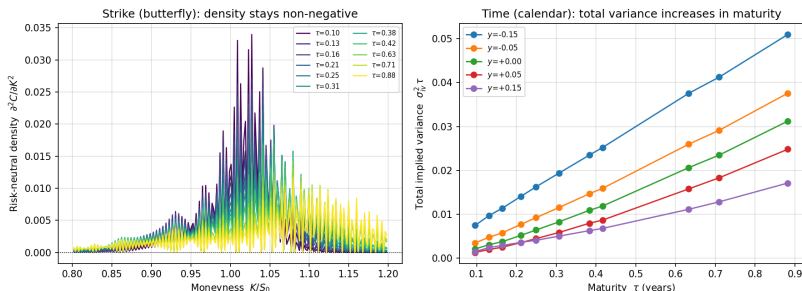
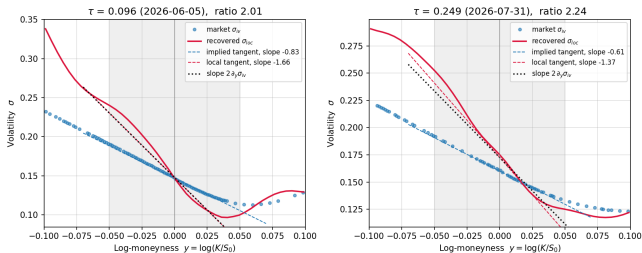


Figure: Left: risk-neutral density $\partial_{KK}^2 \mathcal{C} \geq 0$ per maturity (no butterfly arbitrage). Right: total implied variance $\sigma_{iv}^2 \tau$ increasing in maturity (no calendar arbitrage).

Market surface. The rule of two

Near the money $\partial_y \sigma_{\text{loc}}|_{y=0} = 2 \partial_y \sigma_{\text{iv}}|_{y=0}$. Quadratic fits on $|y| \leq 0.05$ of the market smile and of the calibrated slices:



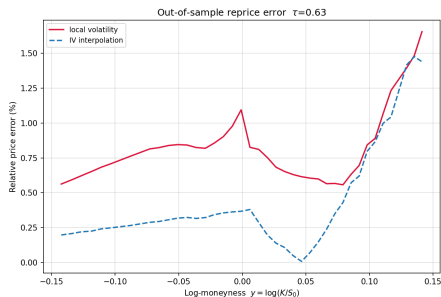
τ	0.096	0.132	0.164	0.211	0.249	0.307	0.384	0.416	0.633	0.710	0.882
ratio	2.01	2.17	2.02	2.36	2.24	2.32	2.19	2.24	2.47	2.17	2.38

The ratio averages 2.16 below three months and stays in $[2.0, 2.5]$: the theory is confirmed empirically.

Market surface. Out of sample pricing

Remove the December 2026 expiration ($\tau = 0.63$) and price it out of sample:

- ▶ **Local volatility**: calibrate on the other ten maturities, the Dupire equation propagates across the gap.
- ▶ **Implied volatility**: interpolate the total variance $\sigma_{iv}^2 \tau$ linearly between $\tau = 0.42$ and 0.71.



Neither is off by more than 2% on $|y| \leq 0.15$. Median error: 0.75% local volatility, 0.29% interpolation. the interpolation is built to reproduce the smile, while the local volatility error accumulates through the PDE. Yet only the local volatility surface is guaranteed to be arbitrage-free.

Conclusions

- ▶ We built the pricing theory from stochastic calculus, through Black-Scholes and arbitrage theory, up to the Dupire forward equation and its no-arbitrage conditions.
- ▶ Calibration of local volatility is an ill-posed inverse problem, and Tikhonov regularization with an adjoint gradient solves it at the cost of two PDE solves per iteration.
- ▶ We reproduced the examples of Egger and Engl: the data determine the volatility near the money, the prior determines the wings.
- ▶ We extended the framework to a time dependent coefficient and calibrated a full SPY surface, arbitrage-free by construction, with sub-percent reprice errors.
- ▶ The rule of two is confirmed empirically on the calibrated surface, and the model reprices a held-out maturity within 2%.

Further work

- ▶ Convergence results for the time dependent extension.
- ▶ Inverse problem formulations for random data, closer to market conditions.

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Thank you